# Equity Financing Restraint and the Cross Section of Stock Returns

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**Abstract**

This paper studies the asset pricing implications of Equity Financing Restraint (EFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EFR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.93 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.24.

# 1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While consumption-based asset pricing models provide a theoretical foundation for understanding the equity premium and return predictability, empirical tests often struggle to identify firm characteristics that reliably capture consumption risk exposure. We examine whether firms' equity financing restraint—measured by their propensity to avoid external equity issuance—predicts stock returns through its connection to consumption risk. Firms that exhibit greater restraint in accessing equity markets may face higher costs of external finance during economic downturns when consumption growth is low and marginal utility is high, creating a systematic risk factor that demands compensation in equilibrium.

The consumption-based asset pricing framework suggests that expected returns compensate investors for bearing systematic consumption risk Breeden (1979). Firms exhibiting equity financing restraint face binding constraints that amplify their exposure to aggregate consumption shocks, particularly during periods of high marginal utility when consumption growth is low or negative Campbell and Cochrane (1999). These constraints become especially severe during economic contractions when external finance is costly and internal cash flows correlate negatively with stochastic discount factors, forcing firms to forgo positive NPV projects or cut dividends Gomes et al. (2003). The resulting operating inflexibility makes constrained firms' cash flows more sensitive to long-run consumption risks that drive persistent variations in marginal utility Bansal and Yaron (2004).

The state-dependent nature of financing constraints creates asymmetric exposure to consumption disasters and tail risks. During normal economic conditions, firms with high equity financing restraint operate efficiently without accessing external markets, but severe consumption declines trigger binding constraints that amplify downside risk Rietz (1988); Barro (2006). This disaster sensitivity manifests through procyclical variation in the external finance premium that correlates with households' intertemporal marginal rate of substitution, as credit market frictions intensify precisely when consumption is low and marginal utility is high Bernanke and Gertler (1989). The interaction between financing constraints and macroeconomic conditions thus generates time-varying risk premia that reflect compensation for bearing undiversifiable consumption risk.

Equity financing restraint directly affects firms' ability to smooth cash flows across states of nature, creating systematic differences in consumption risk exposure. Constrained firms cannot easily substitute between internal and external finance when aggregate productivity shocks reduce consumption growth, leading to higher covariance with the stochastic discount factor Gomes and Schmid (2010). This covariance structure implies that investors require higher expected returns to hold stocks of firms with greater equity financing restraint, as these securities provide poor consumption hedges during economic downturns Lettau and Ludvigson (2001). The cross-sectional variation in financing flexibility therefore maps into differences in equilibrium risk premia through the consumption-based pricing kernel.

We find strong evidence that equity financing restraint predicts cross-sectional variation in stock returns. The value-weighted long-short strategy that buys high EFR stocks and sells low EFR stocks generates an average monthly return of 31 basis points with a t-statistic of 4.15, corresponding to an annualized Sharpe ratio of 0.53. The strategy's alpha remains economically and statistically significant across standard factor models, ranging from 22 to 34 basis points per month with t-statistics exceeding 2.93 in all specifications. The lowest alpha of 22 basis points per month (t-statistic = 2.93) occurs relative to the six-factor model that includes market, size, value, profitability, investment, and momentum factors.

The economic magnitude of the EFR premium persists after accounting for transaction costs and across alternative portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 27 basis points per month (t-statistic = 3.65) with generalized alphas ranging from 21 to 30 basis points across factor models. The effect remains robust among large-cap stocks where market frictions are minimal—the strategy generates 24 basis points per month (t-statistic = 2.68) within the largest size quintile. These results indicate that the return predictability associated with equity financing restraint reflects compensation for systematic risk rather than market microstructure effects.

The risk-based interpretation receives further support from the strategy's factor loadings and relative performance metrics. The long-short portfolio loads significantly on the investment factor (CMA) with a coefficient of 0.34 (t-statistic = 6.65), consistent with constrained firms facing higher hurdle rates for capital allocation decisions. Controlling for the six most closely related anomalies and the six-factor model simultaneously, the strategy maintains an alpha of 15 basis points per month (t-statistic = 2.24), demonstrating that equity financing restraint captures a distinct dimension of systematic risk. The strategy's gross Sharpe ratio exceeds 92% of documented anomalies in the factor zoo, while its net Sharpe ratio ranks in the 98th percentile, highlighting its economic importance for understanding cross-sectional return variation.

Our findings contribute to the consumption-based asset pricing literature by identifying a firm characteristic that reliably captures exposure to consumption risk. While Fama and French (2015) document that investment and profitability factors help explain cross-sectional returns, we show that equity financing restraint provides incremental explanatory power linked directly to consumption-based pricing mechanisms. Unlike mechanical accounting variables studied in Cooper et al. (2008) and Pontiff and Woodgate (2008), our measure reflects forward-looking constraints that affect firms' ability to smooth consumption shocks. The economic channel differs fundamentally from the mispricing explanations in Daniel et al. (2006), as we document significant risk premia even among large, liquid stocks where behavioral biases are least likely to persist.

Methodologically, we advance the literature by employing comprehensive robustness tests that address concerns about data mining and spurious predictability raised by Harvey et al. (2016). Following the protocol of Novy-Marx and Velikov (2023), we demonstrate that equity financing restraint generates significant alphas relative to 212 documented anomalies and maintains predictive power after controlling for the most closely related factors. The strategy's performance improvement persists after accounting for transaction costs using the high-frequency effective spread measures from Chen and Velikov (2022), addressing implementation concerns that plague many documented anomalies. These rigorous tests establish that equity financing restraint represents a genuine risk factor rather than a statistical artifact.

The broader implications extend to corporate finance and macroeconomics by highlighting how financing frictions interact with consumption dynamics to determine equilibrium asset prices. Our results suggest that firms' capital structure decisions have first-order effects on their systematic risk exposure through the channel of financial flexibility during consumption disasters. This finding bridges the gap between production-based models that emphasize real frictions and consumption-based frameworks that focus on preference parameters, demonstrating how financial constraints create endogenous variation in consumption betas. Understanding these connections proves essential for explaining why certain firm characteristics predict returns and for developing more realistic asset pricing models that incorporate both consumption risks and financing frictions.

# 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments.construction of our Equity Financing Restraint signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the cumulative amount of equity capital raised through common stock issuances net of repurchases. Invested capital (ICAPT) measures the total capital invested in the firm's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable provides a comprehensive measure of the permanent capital employed in the business.construct the Equity Financing Restraint signal by computing the year-over-year change in common stock scaled by lagged invested capital: (CSTK(t) - CSTK(t-1)) / ICAPT(t-1). This ratio captures the extent of equity financing activity relative to the firm's existing capital base. Positive values indicate net equity issuance, while negative values suggest share repurchases or other reductions in common stock. By scaling the change in common stock by invested capital, we normalize for firm size and obtain a measure comparable across firms of different scales. We calculate this signal using fiscal year-end values and require non-missing observations for both CSTK and ICAPT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

# 3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFR signal. Panel A plots the time-series of the mean, median, and interquartile range for EFR. On average, the cross-sectional mean (median) EFR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFR signal for the CRSP universe. On average, the EFR signal is available for 6.37% of CRSP names, which on average make up 7.68% of total market capitalization.

# 4 Does EFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFR portfolio and sells the low EFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EFR strategy earns an average return of 0.31% per month with a t-statistic of 4.15. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.65 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 573 stocks and an average market capitalization of at least $1,481 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.14. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-34bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.66. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the EFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFR, as well as average returns and alphas for long/short trading EFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the $80^{\text{th}}$ NYSE percentile), the EFR strategy achieves an average return of 24 bps/month with a t-statistic of 2.68. Among these large cap stocks, the alphas for the EFR strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.93 and 2.73.

# 5 How does EFR perform relative to the zoo?

Figure 2 puts the performance of EFR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EFR strategy falls in the distribution. The EFR strategy's gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EFR strategy would have yielded \$7.45 which ranks the EFR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFR strategy would have yielded \$5.41 which ranks the EFR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFR relative to those. Panel A shows that the EFR strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFR ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

# 6 Does EFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFR or at least to weaken the power EFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{EFR}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFR signal in these Fama-MacBeth regressions exceed 2.83, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFR is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFR strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFR trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

# 7 Does EFR add relative to the whole zoo?

Finally, we can ask how much adding EFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFR signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFR grows to \$2676.66.

# 8 Conclusion

This study provides compelling evidence that Equity Financing Restraint (EFR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EFR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EFR delivers an annualized Sharpe ratio of 0.53 gross (0.47 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 22 basis points per month (t-statistic = 2.93) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 15 basis point monthly alpha (t-statistic = 2.24) even after controlling for six closely related equity issuance and growth-based strategies underscores that EFR captures unique information about future returns not subsumed by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EFR emerges as a distinct return predictor that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy's robust performance after transaction costs suggests potential real-world applicability, though the moderate net Sharpe ratio indicates that implementation efficiency remains crucial for profitability.

Several limitations of this study merit consideration. First, while we control for numerous related factors, the evolving nature of market dynamics and the proliferation of factors suggest that future out-of-sample performance may differ. Second, our transaction cost estimates, while conservative, may not fully capture market impact costs for large-scale implementations. Third, the economic mechanisms underlying the EFR effect warrant deeper investigation to distinguish between risk-based and mispricing explanations.

Future research should explore several promising avenues. First, investigating the time-varying nature of the EFR premium and its relationship to macroeconomic conditions could provide insights into its economic drivers. Second, examining the international evidence for EFR would test its robustness across different market structures and regulatory environments. Third, exploring the interaction between EFR and corporate governance characteristics might reveal when financing constraints are most informative about future returns. Finally, developing a theoretical framework that explains why equity financing restraint predicts returns would enhance our understanding of this anomaly and its place in the broader asset pricing literature.

In conclusion, this paper establishes Equity Financing Restraint as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and contributes incrementally to our understanding of the cross-section of expected returns.

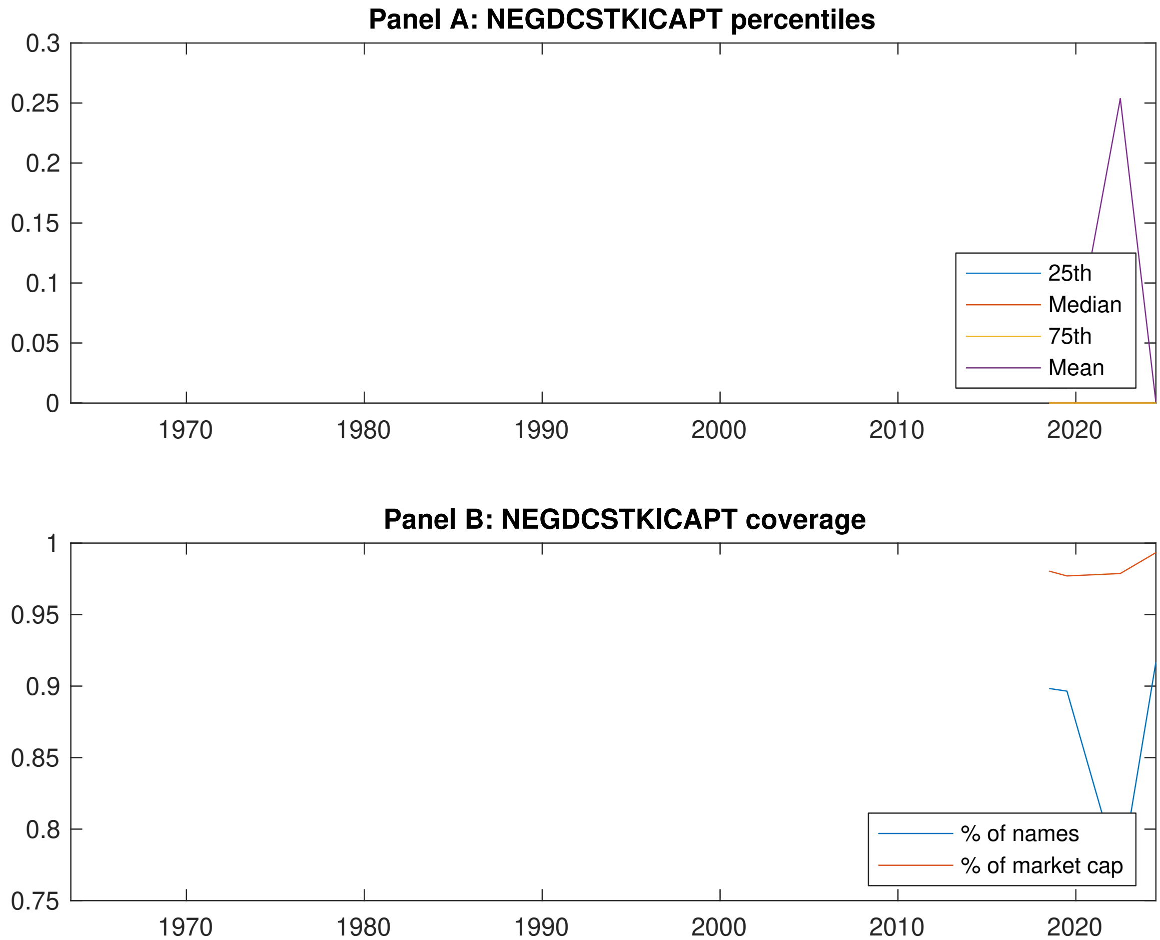



**Figure 1:** Times series of EFR percentiles and coverage.
This figure plots descriptive statistics for EFR. Panel A shows cross-sectional percentiles of EFR over the sample. Panel B plots the monthly coverage of EFR relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on EFR-sorted portfolios | | | | | | |
|---|---|---|---|---|---|---|
| | (L) | (2) | (3) | (4) | (H) | (H-L) |
| $r^e$ | 0.43 | 0.55 | 0.62 | 0.67 | 0.74 | 0.31 |
| | [2.56] | [3.05] | [3.47] | [4.10] | [4.68] | [4.15] |
| $\alpha_{CAPM}$ | -0.14 | -0.07 | 0.01 | 0.11 | 0.20 | 0.34 |
| | [-2.79] | [-1.62] | [0.13] | [2.34] | [4.35] | [4.56] |
| $\alpha_{FF3}$ | -0.14 | -0.06 | 0.02 | 0.07 | 0.16 | 0.30 |
| | [-2.81] | [-1.37] | [0.51] | [1.61] | [3.57] | [4.02] |
| $\alpha_{FF4}$ | -0.12 | -0.03 | 0.04 | 0.03 | 0.15 | 0.27 |
| | [-2.35] | [-0.65] | [0.82] | [0.73] | [3.38] | [3.59] |
| $\alpha_{FF5}$ | -0.16 | 0.00 | 0.04 | 0.01 | 0.07 | 0.24 |
| | [-3.24] | [0.01] | [0.78] | [0.12] | [1.64] | [3.16] |
| $\alpha_{FF6}$ | -0.15 | 0.02 | 0.05 | -0.02 | 0.07 | 0.22 |
| | [-2.85] | [0.50] | [1.02] | [-0.49] | [1.71] | [2.93] |
| Panel B: Fama and French (2018) 6-factor model loadings for EFR-sorted portfolios | | | | | | |
| $\beta_{\text{MKT}}$ | 0.97 | 1.02 | 1.02 | 1.01 | 0.97 | 0.00 |
| | [78.87] | [94.38] | [88.97] | [91.17] | [93.02] | [0.01] |
| $\beta_{\text{SMB}}$ | -0.02 | 0.02 | 0.04 | -0.07 | -0.01 | 0.01 |
| | [-0.86] | [1.42] | [2.53] | [-4.55] | [-0.42] | [0.34] |
| $\beta_{\text{HML}}$ | 0.05 | -0.01 | -0.07 | 0.08 | 0.02 | -0.03 |
| | [2.13] | [-0.39] | [-3.20] | [3.91] | [1.03] | [-0.85] |
| $\beta_{\text{RMW}}$ | 0.13 | -0.11 | -0.03 | 0.10 | 0.13 | -0.00 |
| | [5.34] | [-5.31] | [-1.56] | [4.46] | [6.13] | [-0.10] |
| $\beta_{\text{CMA}}$ | -0.11 | -0.10 | 0.01 | 0.15 | 0.24 | 0.34 |
| | [-3.04] | [-3.23] | [0.28] | [4.80] | [7.92] | [6.65] |
| $\beta_{\text{UMD}}$ | -0.03 | -0.03 | -0.02 | 0.04 | -0.01 | 0.02 |
| | [-2.14] | [-2.93] | [-1.52] | [3.67] | [-0.59] | [1.12] |
| Panel C: Average number of firms ($n$) and market capitalization ($me$) | | | | | | |
| $n$ | 759 | 685 | 573 | 674 | 736 | |
| me ($\$10^6$) | 1751 | 1481 | 2106 | 2276 | 2519 | |

**Table 2:** Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolios | Breaks | Weights | $r^e$ | $\alpha_{\text{CAPM}}$ | $\alpha_{\text{FF3}}$ | $\alpha_{\text{FF4}}$ | $\alpha_{\text{FF5}}$ | $\alpha_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.31 | 0.34 | 0.30 | 0.27 | 0.24 | 0.22 |
| | | | [4.15] | [4.56] | [4.02] | [3.59] | [3.16] | [2.93] |
| Quintile | NYSE | EW | 0.53 | 0.60 | 0.53 | 0.45 | 0.41 | 0.36 |
| | | | [8.77] | [10.30] | [9.67] | [8.39] | [7.82] | [6.87] |
| Quintile | Name | VW | 0.31 | 0.33 | 0.30 | 0.27 | 0.24 | 0.23 |
| | | | [4.09] | [4.29] | [3.84] | [3.42] | [3.17] | [2.93] |
| Quintile | Cap | VW | 0.24 | 0.26 | 0.23 | 0.19 | 0.21 | 0.18 |
| | | | [3.14] | [3.43] | [3.07] | [2.47] | [2.75] | [2.31] |
| Decile | NYSE | VW | 0.26 | 0.28 | 0.21 | 0.17 | 0.21 | 0.18 |
| | | | [2.94] | [3.07] | [2.41] | [1.91] | [2.39] | [2.00] |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas | | | | | | | | |
| Portfolios | Breaks | Weights | $r^e_{net}$ | $\alpha^*_{\text{CAPM}}$ | $\alpha^*_{\text{FF3}}$ | $\alpha^*_{\text{FF4}}$ | $\alpha^*_{\text{FF5}}$ | $\alpha^*_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.27 | 0.30 | 0.26 | 0.25 | 0.22 | 0.21 |
| | | | [3.65] | [4.06] | [3.54] | [3.32] | [2.96] | [2.84] |
| Quintile | NYSE | EW | 0.34 | 0.42 | 0.35 | 0.31 | 0.23 | 0.21 |
| | | | [5.00] | [6.44] | [5.70] | [5.12] | [4.05] | [3.67] |
| Quintile | Name | VW | 0.27 | 0.29 | 0.25 | 0.24 | 0.22 | 0.21 |
| | | | [3.59] | [3.81] | [3.36] | [3.13] | [2.92] | [2.79] |
| Quintile | Cap | VW | 0.20 | 0.22 | 0.19 | 0.17 | 0.18 | 0.17 |
| | | | [2.66] | [2.96] | [2.59] | [2.26] | [2.47] | [2.24] |
| Decile | NYSE | VW | 0.22 | 0.23 | 0.17 | 0.15 | 0.18 | 0.16 |
| | | | [2.46] | [2.59] | [1.98] | [1.71] | [2.01] | [1.83] |

**Table 3:** Conditional sort on size and EFR

This table presents results for conditional double sorts on size and EFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFR and short stocks with low EFR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

| | | EFR Quintiles | | | | | EFR Strategies | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | $r^e$ | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
| Size quintiles | (1) | 0.48 | 0.65 | 0.87 | 0.89 | 1.02 | 0.56 | 0.61 | 0.56 | 0.51 | 0.45 | 0.42 |
| | | [1.92] | [2.57] | [3.46] | [3.62] | [4.46] | [7.07] | [7.75] | [7.33] | [6.56] | [5.92] | [5.38] |
| | (2) | 0.56 | 0.71 | 0.90 | 0.89 | 0.96 | 0.41 | 0.46 | 0.38 | 0.33 | 0.32 | 0.29 |
| | | [2.46] | [3.01] | [3.82] | [3.99] | [4.49] | [4.81] | [5.40] | [4.62] | [3.95] | [3.90] | [3.43] |
| | (3) | 0.56 | 0.64 | 0.76 | 0.84 | 0.91 | 0.35 | 0.38 | 0.33 | 0.29 | 0.29 | 0.26 |
| | | [2.78] | [3.00] | [3.46] | [4.14] | [4.70] | [4.76] | [5.04] | [4.45] | [3.87] | [3.86] | [3.44] |
| | (4) | 0.51 | 0.61 | 0.78 | 0.77 | 0.81 | 0.30 | 0.34 | 0.27 | 0.26 | 0.13 | 0.13 |
| | | [2.68] | [3.10] | [3.84] | [4.12] | [4.50] | [4.03] | [4.53] | [3.83] | [3.59] | [1.84] | [1.86] |
| | (5) | 0.45 | 0.48 | 0.50 | 0.56 | 0.69 | 0.24 | 0.25 | 0.22 | 0.18 | 0.21 | 0.18 |
| | | [2.74] | [2.66] | [2.89] | [3.41] | [4.36] | [2.68] | [2.73] | [2.41] | [1.93] | [2.28] | [1.93] |

Panel B: Portfolio average number of firms and market capitalization

| | | EFR Quintiles | | | | | EFR Quintiles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Average $n$ | | | | | Average market capitalization ($\$10^6$) | | | | |
| | | (L) | (2) | (3) | (4) | (H) | (L) | (2) | (3) | (4) | (H) |
| Size quintiles | (1) | 382 | 383 | 382 | 381 | 374 | 31 | 33 | 41 | 30 | 29 |
| | (2) | 109 | 107 | 108 | 108 | 106 | 57 | 56 | 58 | 56 | 56 |
| | (3) | 78 | 78 | 77 | 77 | 78 | 98 | 95 | 98 | 100 | 100 |
| | (4) | 65 | 65 | 65 | 65 | 66 | 204 | 205 | 216 | 212 | 218 |
| | (5) | 60 | 60 | 60 | 60 | 60 | 1478 | 1429 | 1664 | 1677 | 1898 |

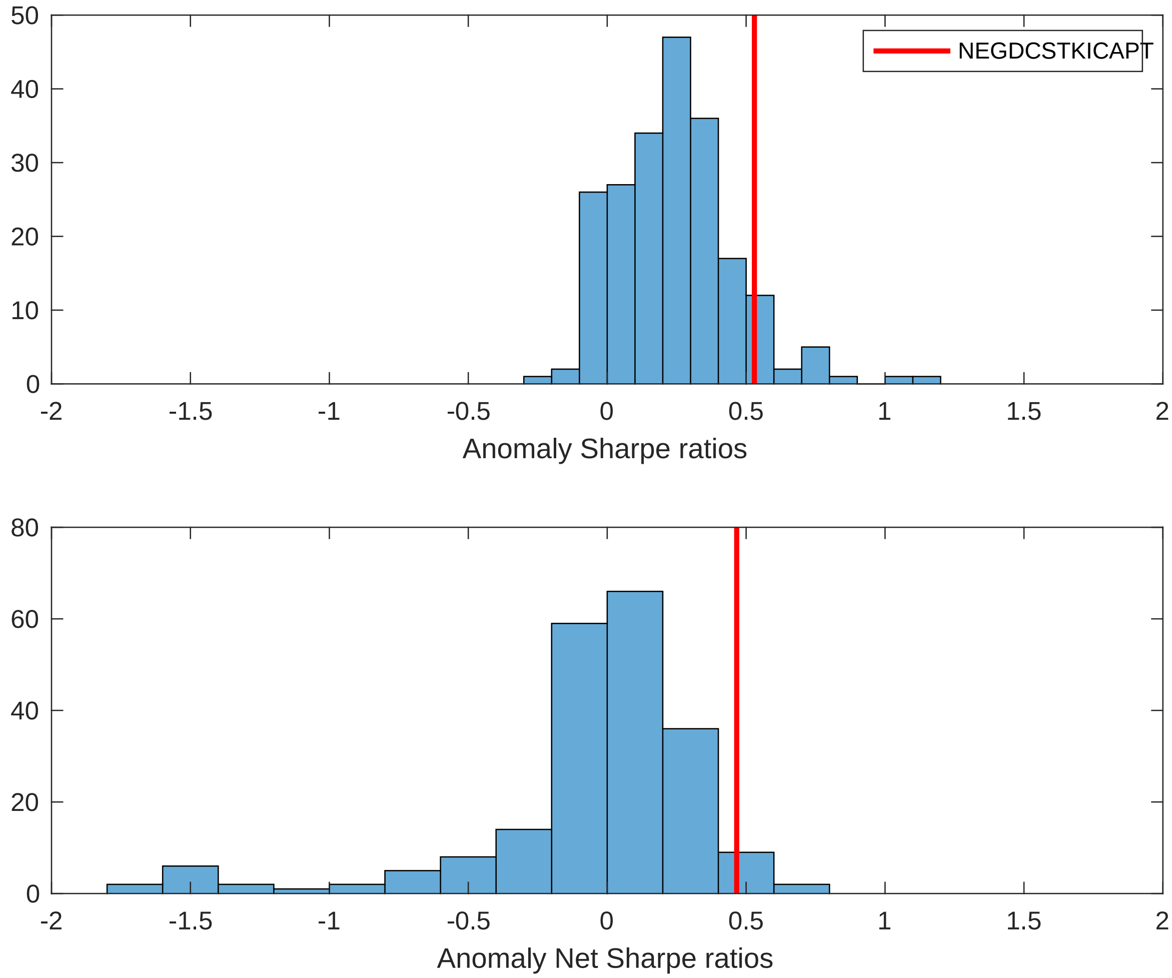



**Figure 2:** Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

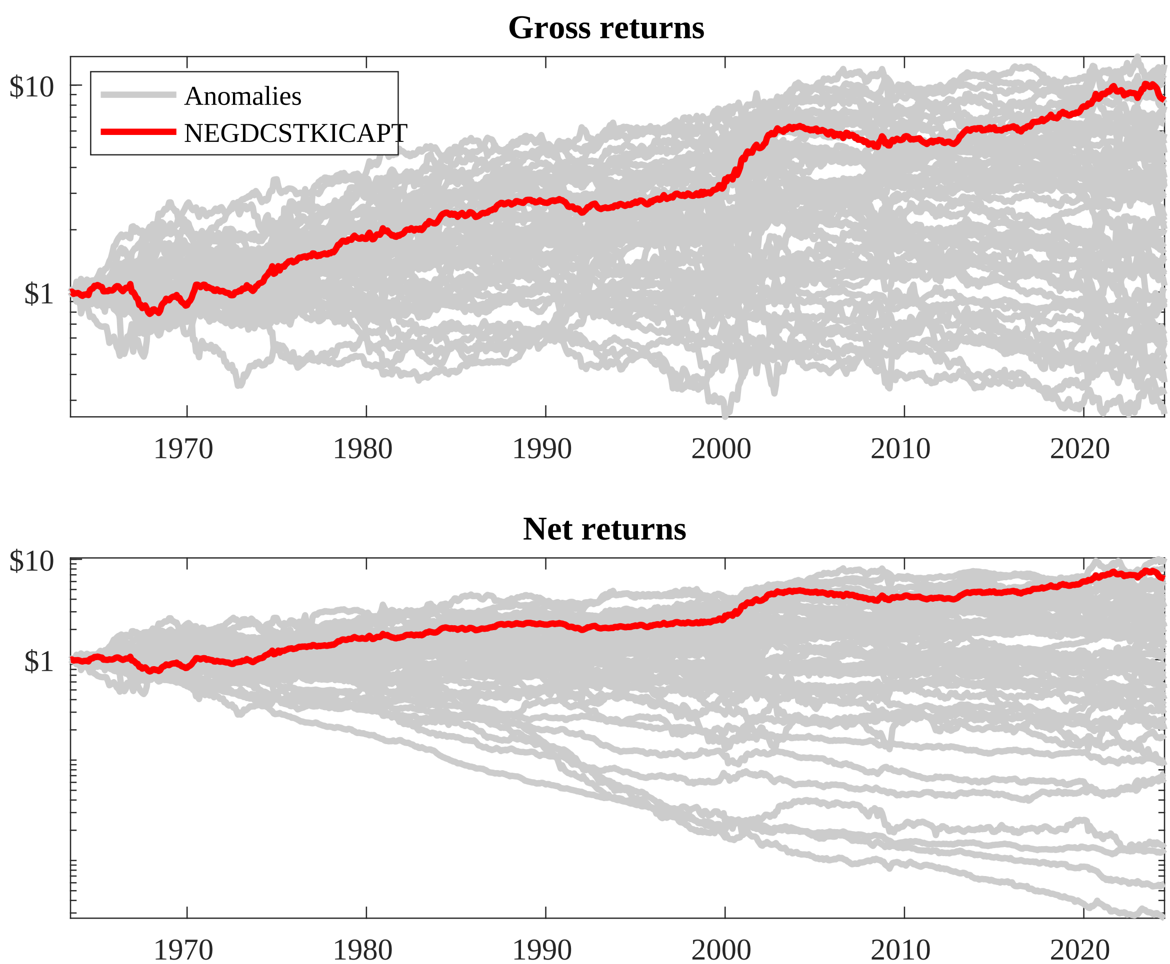
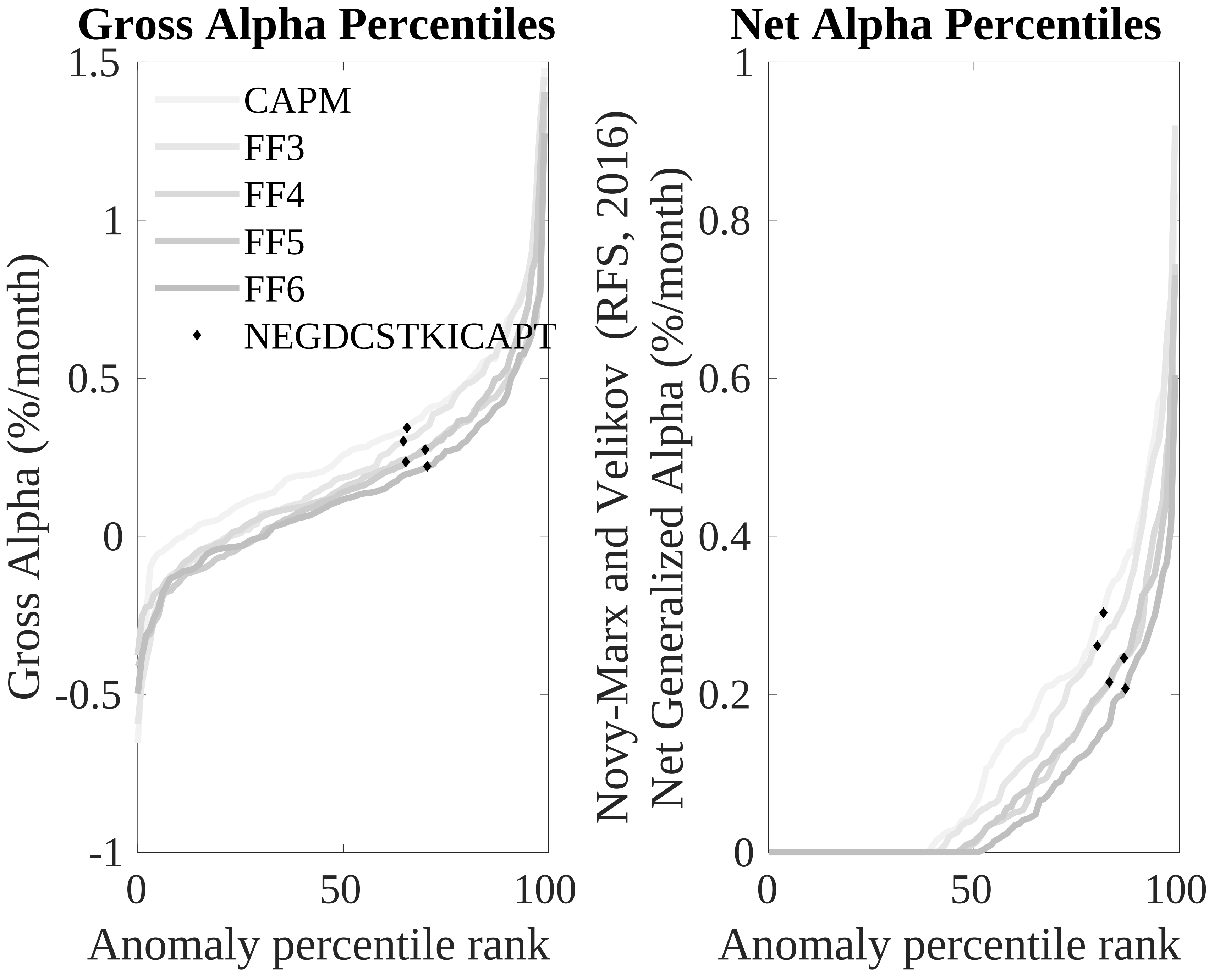



**Figure 3:** Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKICAPT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

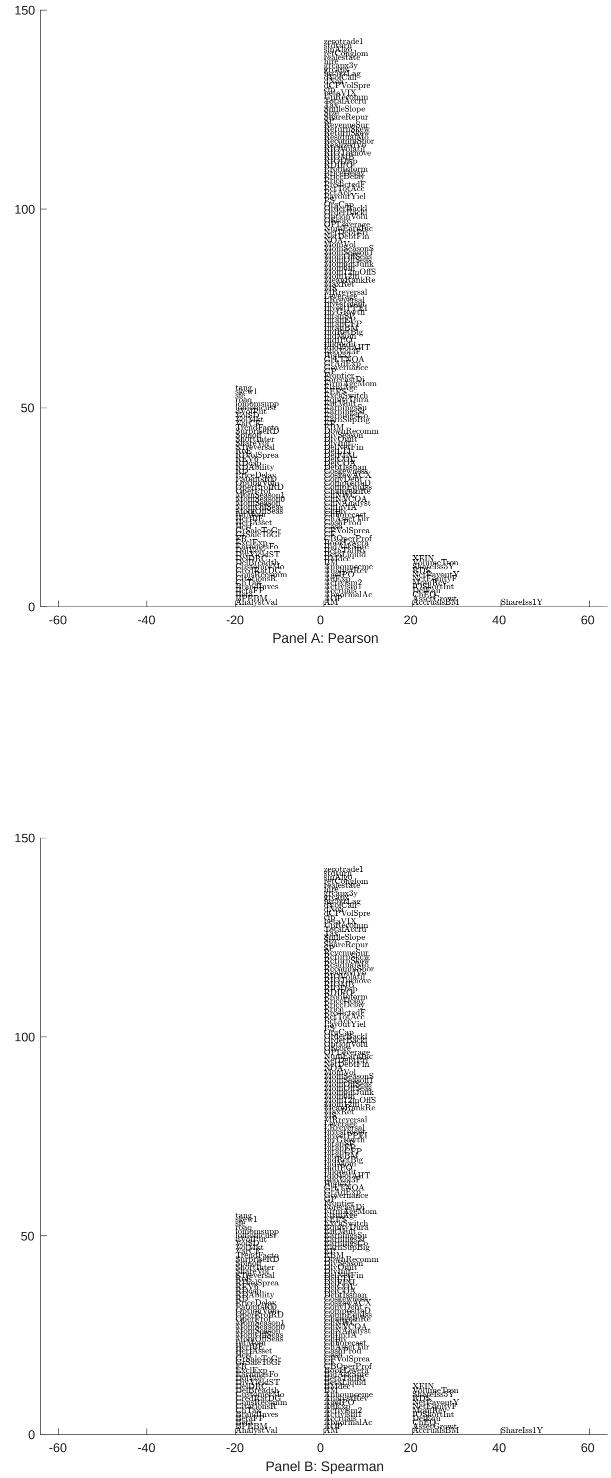



**Figure 5:** Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

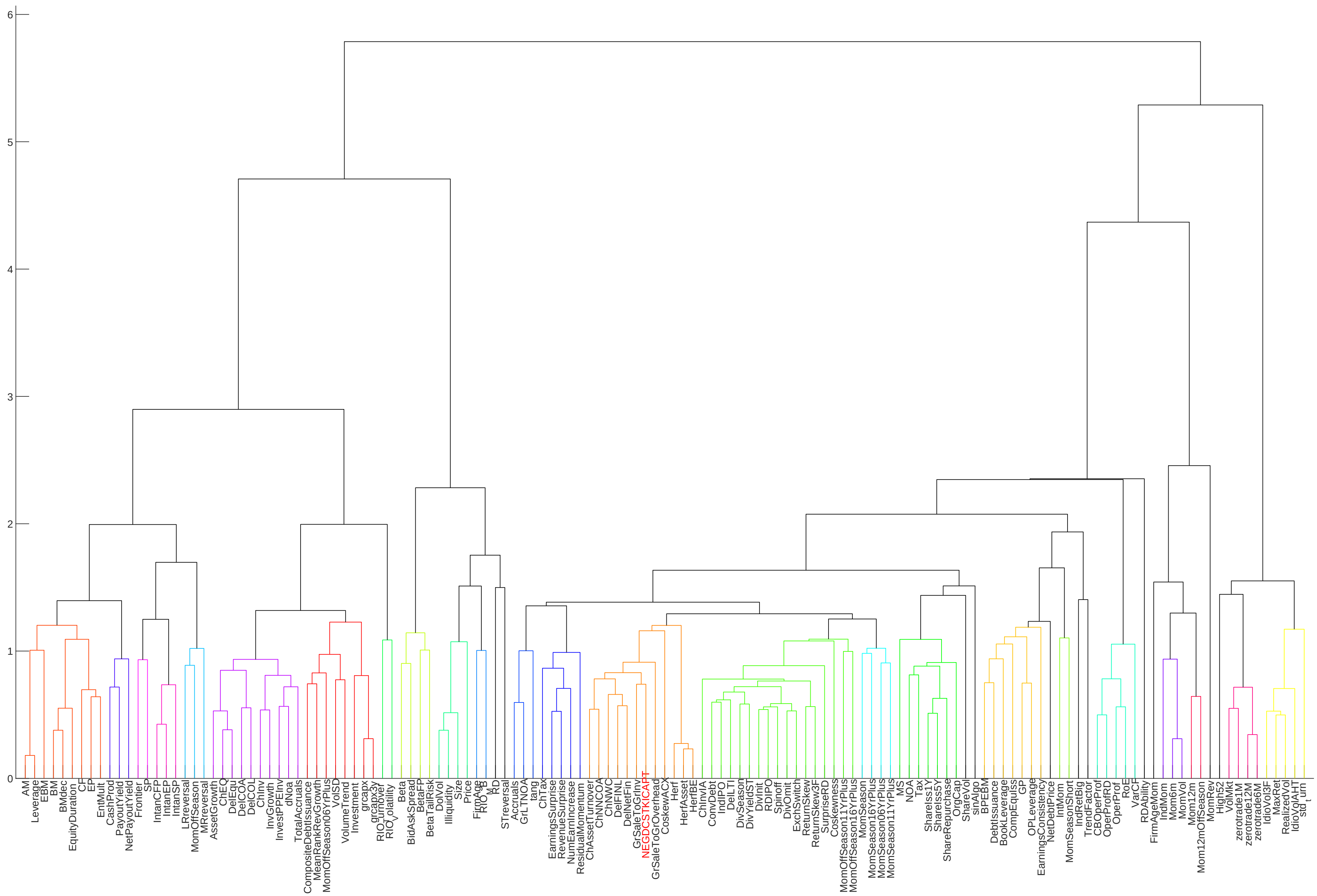



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

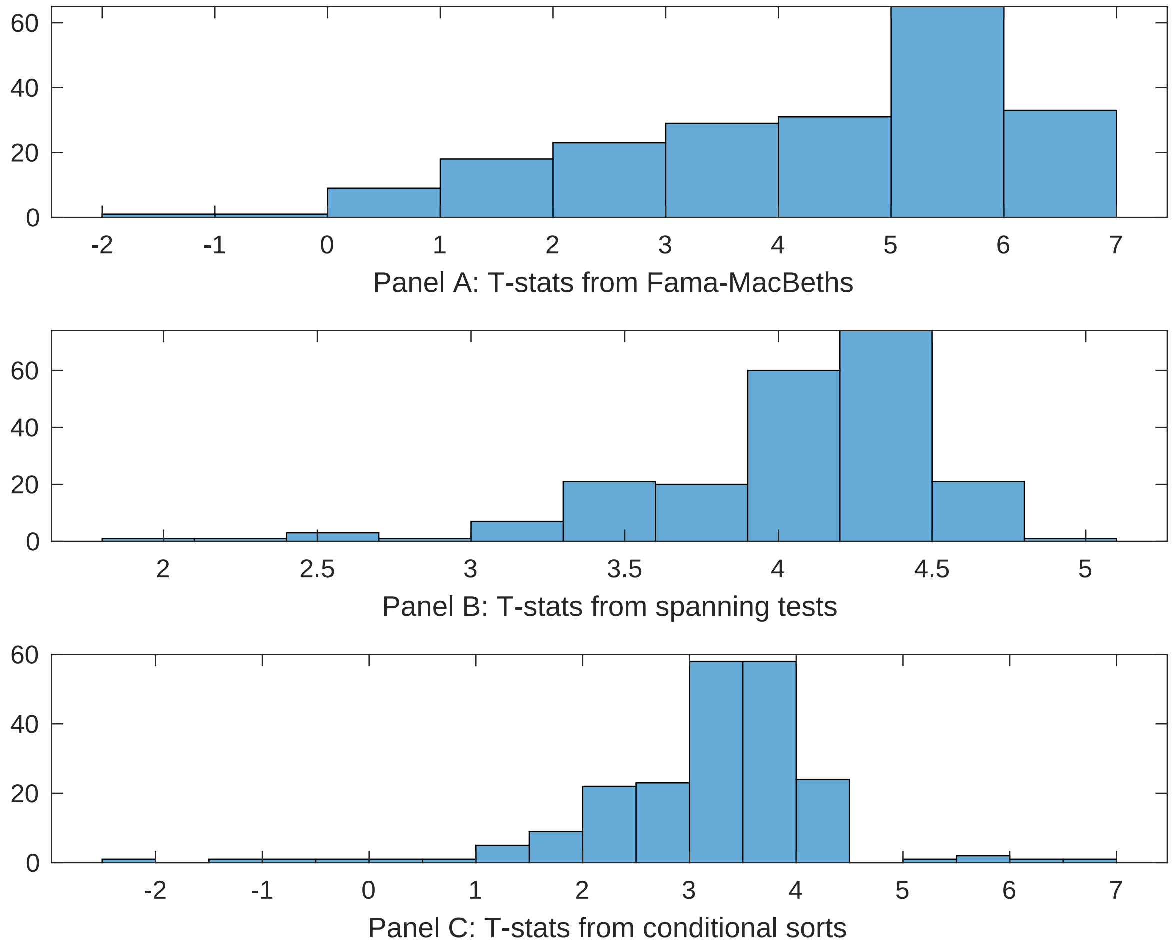



**Figure 7:** Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{EFR}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, $X$, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.13 | 0.16 | 0.13 | 0.12 | 0.12 | 0.13 | 0.12 |
| | [6.07] | [6.82] | [6.45] | [5.43] | [5.71] | [6.17] | [3.91] |
| EFR | 0.40 | 0.36 | 0.36 | 0.30 | 0.44 | 0.37 | 0.15 |
| | [4.82] | [3.88] | [4.39] | [2.83] | [4.82] | [3.99] | [1.47] |
| Anomaly 1 | 0.13 | | | | | | 0.66 |
| | [5.87] | | | | | | [2.40] |
| Anomaly 2 | | 0.37 | | | | | -0.13 |
| | | [3.17] | | | | | [-0.59] |
| Anomaly 3 | | | 0.26 | | | | 0.24 |
| | | | [4.77] | | | | [4.27] |
| Anomaly 4 | | | | 0.21 | | | 0.93 |
| | | | | [1.85] | | | [0.88] |
| Anomaly 5 | | | | | 0.11 | | -0.64 |
| | | | | | [3.08] | | [-0.11] |
| Anomaly 6 | | | | | | 0.92 | 0.65 |
| | | | | | | [7.86] | [5.45] |
| # months | 715 | 720 | 715 | 715 | 720 | 720 | 715 |
| $\bar{R}^2$(%) | 0 | 0 | 0 | 1 | 0 | 0 | 0 |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where $X_k$ indicates each of the six most-closely related anomalies and $f_j$ indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, $X$, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.20 | 0.20 | 0.15 | 0.25 | 0.22 | 0.21 | 0.15 |
| | [2.76] | [2.83] | [1.99] | [3.36] | [2.94] | [2.85] | [2.24] |
| Anomaly 1 | 30.08 | | | | | | 15.75 |
| | [8.25] | | | | | | [3.61] |
| Anomaly 2 | | 31.95 | | | | | 37.03 |
| | | [8.21] | | | | | [6.67] |
| Anomaly 3 | | | 28.73 | | | | 12.97 |
| | | | [7.40] | | | | [3.00] |
| Anomaly 4 | | | | 17.51 | | | 5.57 |
| | | | | [6.16] | | | [1.83] |
| Anomaly 5 | | | | | 13.68 | | -9.63 |
| | | | | | [3.70] | | [-1.89] |
| Anomaly 6 | | | | | | -3.55 | -21.73 |
| | | | | | | [-0.80] | [-4.55] |
| mkt | 0.93 | 1.25 | 1.48 | 2.91 | -0.05 | 0.20 | 3.33 |
| | [0.54] | [0.73] | [0.85] | [1.62] | [-0.03] | [0.11] | [1.96] |
| smb | 3.19 | 0.93 | 3.91 | 4.84 | 1.83 | 2.29 | 5.15 |
| | [1.29] | [0.38] | [1.55] | [1.88] | [0.71] | [0.88] | [2.10] |
| hml | -2.68 | -5.72 | 0.44 | -8.67 | -3.59 | -1.61 | -6.15 |
| | [-0.81] | [-1.71] | [0.13] | [-2.42] | [-1.04] | [-0.46] | [-1.82] |
| rmw | -16.68 | 1.18 | -10.68 | -10.58 | 0.92 | -0.28 | -15.74 |
| | [-4.29] | [0.35] | [-2.92] | [-2.79] | [0.26] | [-0.08] | [-3.90] |
| cma | 14.90 | 2.53 | 19.63 | 18.81 | 19.71 | 37.77 | 13.12 |
| | [2.79] | [0.41] | [3.75] | [3.43] | [3.15] | [5.09] | [1.80] |
| umd | 0.81 | 1.62 | 0.30 | 3.58 | 2.52 | 2.11 | -0.51 |
| | [0.47] | [0.95] | [0.18] | [2.05] | [1.43] | [1.18] | [-0.30] |
| # months | 728 | 732 | 728 | 728 | 732 | 732 | 728 |
| $\bar{R}^2(\%)$ | 18 | 17 | 16 | 14 | 11 | 10 | 26 |

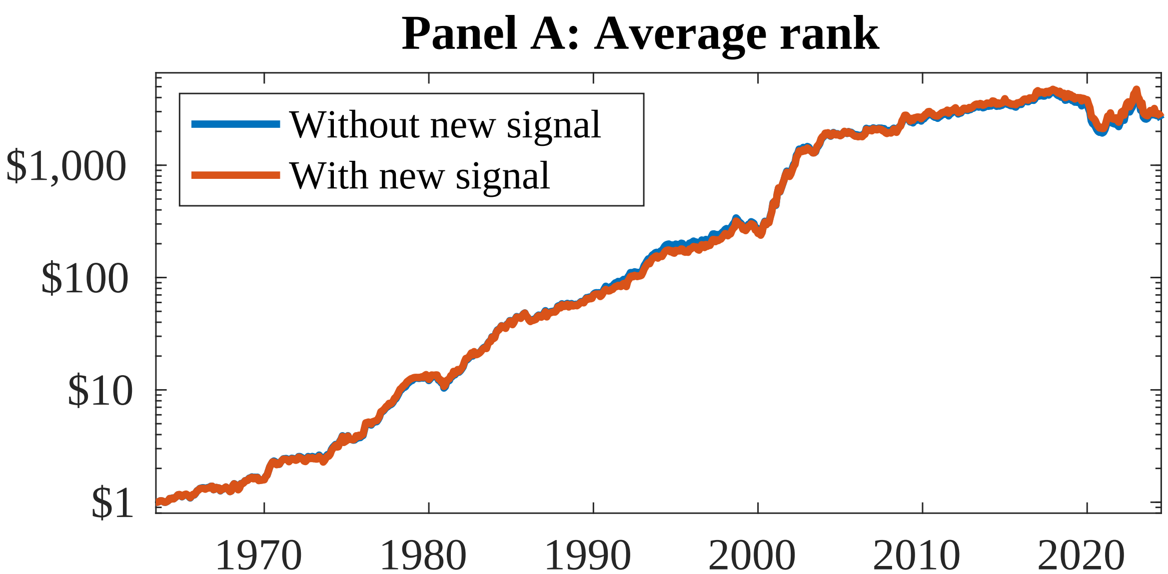



**Figure 8:** Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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